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Case Study 3 : AI for Cross-Culture Learning

Project Title : ASEAN Culture Guide AI (SeaAI)

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1.0 Introduction

1.1 About ASEAN Culture Guide AI (SeaAI)

ASEAN Culture Guide AI (SeaAI) is designed to bridge cross-cultural communication and educational gaps within the regions of Southeast Asian countries. Presently, language barriers, diverse business practices, divergence in social, cultural as well as religious values, and over-relianced on localised national history may lead to costly misunderstandings and slow regional integration. SeaAI assists addressing these and many more challenges by acting as an AI-powered cultural bridge or guide. Our solution directly aligns with the UN's 2030 Sustainable Development Goals: SDG 4 (Quality Education), SDG 10 (Reduced Inequalities), and SDG 11 (Sustainable Cities and Communities).

1.2 Factors : Why is it a problem?

In the eastern hemispheres bordered by the Indian and Pacific Oceans, Southeast Asia is a subregion of Asia that comprises 11 countries, encompassing hundreds of distinct languages, religions and cultural heritages. While diversity is a regional strength in the area, it also generates significant friction for integration between groups of people. As cross-border mobility increases through factors such as international youth exchange programmes, tourism, international migrants, and multinational economic practices, citizens are constantly exposed to unfamiliar social norms. In hindsight, the absence of a centralised and accessible tool to understand these diverse business practices, social etiquettes as well as implicit rules in everyday interactions would only frequently result in miscommunications. This ultimately hinders both educational and organisational collaboration while also slows down economic partnerships across ASEAN countries.

1.3 Target Audience

The target audience for SeaAI is specifically engineered for three primary groups: students, business professionals, and tourists. These demographics are essentially known for having to experience high-frequency interactions with heterogeneous backgrounds daily as they require immediate context-aware assistance the most.

1.4 Objectives : Main Goal

Our primary objective for ASEAN Culture Guide AI (SeaAI) is to bridge the identity and communication gap between ASEAN citizens, as well as between locals and Non-ASEAN visitors. Smoother cross-border partnerships, strengthened regional knowledge and fostered empathy can be built through SeaAI by providing instant cultural context.

2.0 Review on Existing Solutions

2.1 Example of Previous Solutions

Cross-culture Learning has been made easier with the use of modern technology and especially so with the usage of Artificial Intelligence (AI) into applications and websites of these pre-existing solutions in bridging the gap for cross-culture learning. Apps with AI integration such as ChatGPT, Google Gemini, ChanceAi and FaceMap are the few mainstream tools in learning as well as exploring new cultures of a region or country.

2.2 Problem and Weaknesses of Pre-existing Solutions

However, weaknesses are able to occur within these apps that might cause problems for users in terms of inaccuracy or lack of information especially in lesser known cultures due to Western-centric bias, inability to understand and comprehend users questions or error in image detection.

Advanced AI applications such as ChatGPT and Google Gemini are highly developed, equipped with multimodal capabilities and vast knowledge bases that can also solve complex tasks but sometimes their responses can be faulty due to lack of information or miscommunication in interpreting users' needs. For example, ChatGPT would sometimes hallucinate incorrect facts and deviate from the original question while Gemini sometimes have errors analysing uploaded pictures and would also make assumptions due to lack of information. Moreover, both apps have permanent storage limits in terms of uploading files or pictures and are also locked behind a paywall which may prove to be inconvenient for learning students or tourists. These weaknesses can both hinder user experience, time efficiency and knowledge accuracy.

2.3 Differences With SeaAI with other AI solutions

In our ASEAN Culture Guide Ai (SeaAI), the difference relies on where the website itself, unlike general-purpose AIs, is purely focused on providing hyper-local learning of cultures in ASEAN countries where its people have diverse traditions and norms. In addition, SeaAI will provide as many efficient and necessary tools as our market is primarily targeted towards students and travelers which are individuals known to be cost-conscious.

3.0 Methodology

3.1 Prototype Development & Tech Stack

The development of ASEAN Culture Guide AI (SeaAI) started with creating a working prototype to show how the system could help people learn about different cultures across ASEAN. The prototype acts as a hands-on demonstration solution for making cross-cultural learning easier and more interactive.

The prototype is built using Google AI Studio, which allows us to quickly design and test AI applications. At the heart of the system is the Gemini AI model by Google, which helps interpret user questions and provide clear, relevant cultural responses.

The interface was designed to be simple and easy to use. Users can ask questions, upload images of cultural items like traditional food or clothing, and receive explanations about their significance. The prototype includes key features such as a cultural guide, an AI assistant, and an image recognition tool that brings cultural learning to life.

Finally, we tested the prototype by simulating real user interactions to make sure the AI gave helpful and accurate information, and that the platform was easy to navigate. This prototype serves as a proof-of-concept, showing how AI can help people better understand and respect the diverse cultures of the ASEAN region.

3.2 System Architecture & Feature Flow

SeaAI was developed as a solution to address the challenges of cross-cultural misunderstanding and learning gaps in ASEAN countries. The solution focuses on providing users with practical, real-time information to navigate social, cultural, and business interactions respectfully.

The platform combines multiple features to deliver a complete cultural guidance system:

1. Digital Cultural Encyclopedia – Provides users with accessible information about each ASEAN country's traditions, etiquette, and local norms.

2. Interactive AI Assistance – Users can chat with the AI assistant to ask questions about cultural practices, plan responsible trips, or learn basic phrases in local languages.

3. Image Recognition for Cultural Items – Users can capture photos of food, clothing, or artifacts, and the AI identifies and explains their cultural and historical relevance.

4. Inclusive and Sustainable Engagement – The system encourages practices that support local communities, preserves cultural heritage, and promotes sustainable tourism.

The solution is designed to be user-friendly, cost-conscious, and focused on the unique needs of students, travelers, and professionals. By combining these features, SeaAI offers a practical and scalable approach to fostering cultural understanding across the ASEAN region.

4.0 Results

4.1 Objectives Fulfilled : How it solves the Problem

By using SeaAI, it helps to reduce communication barriers among ASEAN countries. In our website, it provides information about language, and common phrases for each ASEAN country. This helps users to communicate better, reduce misunderstanding during meetings, collaborations, and any other discussions.

Second, it improves awareness of different cultural values and social norms. Through categories such as People and Ethnicity, Culture and Customs, and Dance and Arts, users gain a clearer understanding of traditions and behaviours that are considered important, respectful or appropriate in each country. This knowledge helps avoid confusion during negotiations or cultural exchanges.

Next, SeaAI also helps users understand different work and business practices across ASEAN. By learning about cultural expectations, communication styles, and social etiquette more effectively and building strong regional partnerships.

Finally, SeaAI promotes regional awareness and shared ASEAN identity. Since traditional education systems focus mainly on national history, the platform provides an accessible way for users to learn about neighbouring ASEAN cultures. Using SeaAI encourages greater appreciation of diversity while strengthening the understanding of shared regional values.

Overall, SeaAI fulfills its objectives by providing a user-friendly platform that enhances cross-cultural understanding, improves communication and supports stronger cooperation among ASEAN communities, directly advancing SDG 4 (Quality Education) and SDG 10 (Reduced Inequalities) by democratizing access to cultural knowledge.

4.2 Marketability: How does the app survive

First, SeaAI has high market potential among university students, especially those involved in ASEAN student exchange programs. The platform helps students understand cultural differences, improve communication skills, and adapt more easily when studying in other ASEAN countries.

Second, it also has high market potential among business professionals in ASEAN. The platform helps businesspeople understand regional differences in culture, religion, and work practices, allowing them to build stronger business relationships and partnerships.

Third, SeaAI has potential among travelers and tourists visiting ASEAN countries. Travelers that are interested in learning about local customs, food, language, and culture can use SeaAI before visiting a country in ASEAN. SeaAI can serve as a quick cultural guide that helps travelers understand local traditions, and have more meaningful travel experiences while exploring different ASEAN destinations.

To ensure long-term commercial viability and fund continuous refinements and optimisation, a "Freemium model" will be introduced where users will receive basic standardised text translations and images for free but must pay a small fee to unlock premium features such as unlimited AI cultural chats and advanced image recognition. A B2B (Business-to-Business) subscription model will also be provided to large institutions such as universities and large corporations where they will pay an annual fee to give their students or workers the premium access to SeaAI.

5.0 Output and UI Screenshots of Prototype

Figure 1

Figure 2

Figure 1 & Figure 2 : The SeaAI user interface and regional selection dashboard

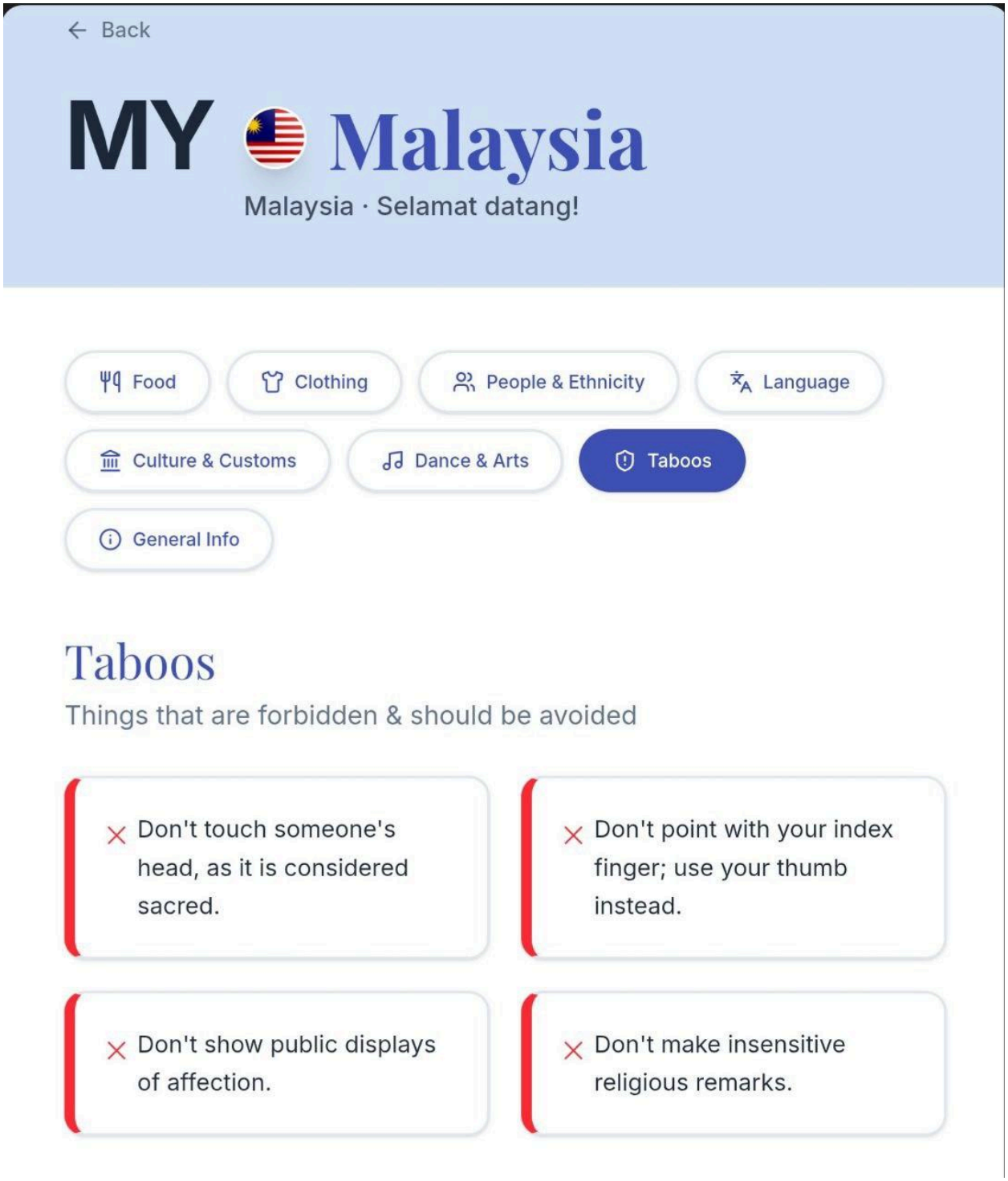
Back
MY
Malaysia
Malaysia · Selamat datang!
Food
Clothing
People & Ethnicity
Language
Culture & Customs
Dance & Arts
Taboos
General Info
Taboos
Things that are forbidden & should be avoided
Don't touch someone's head, as it is considered sacred.
Don't point with your index finger; use your thumb instead.
Don't show public displays of affection.
Don't make insensitive religious remarks.

Figure 3

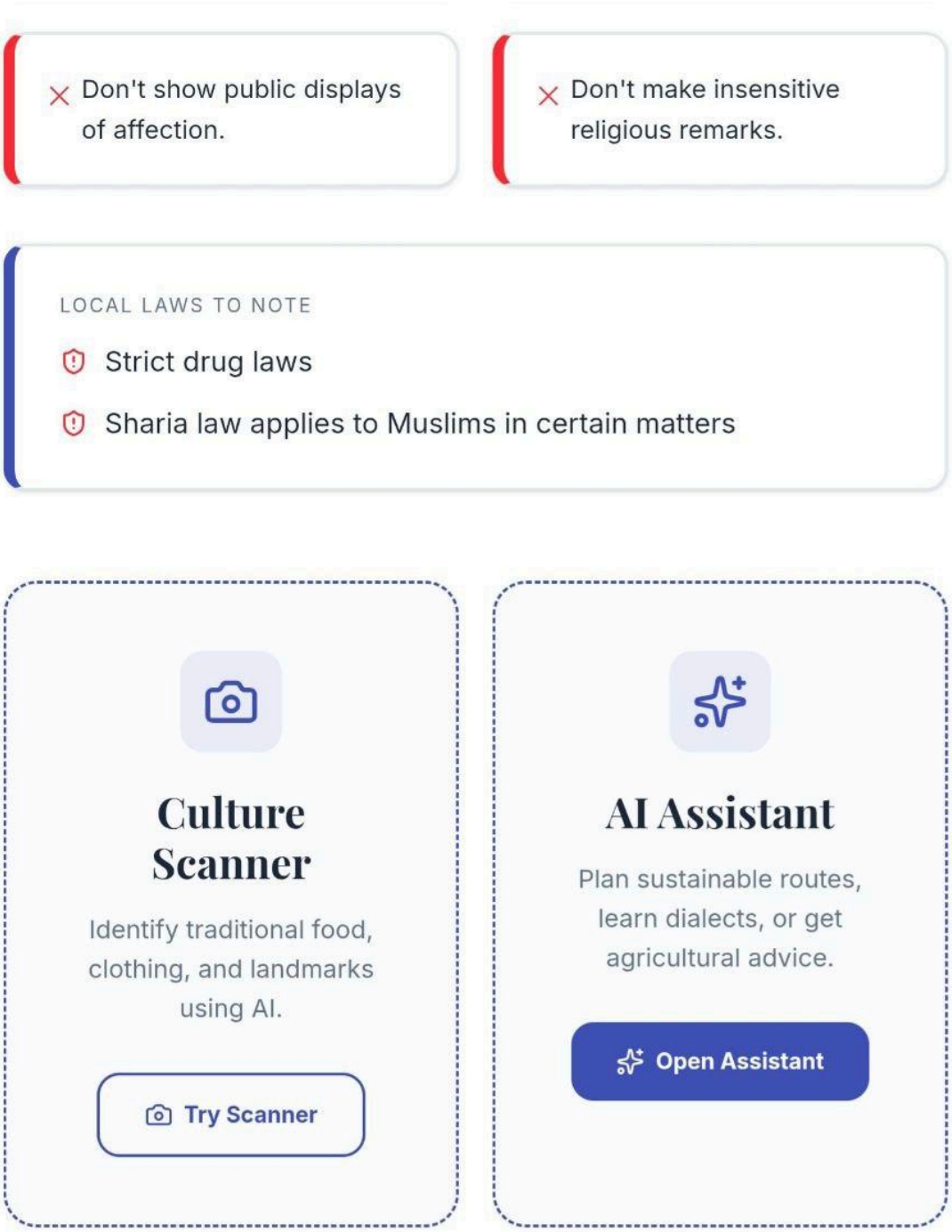

Figure 4

Figure 3 & Figure 4: ASEAN country basic knowledge categorisation interface

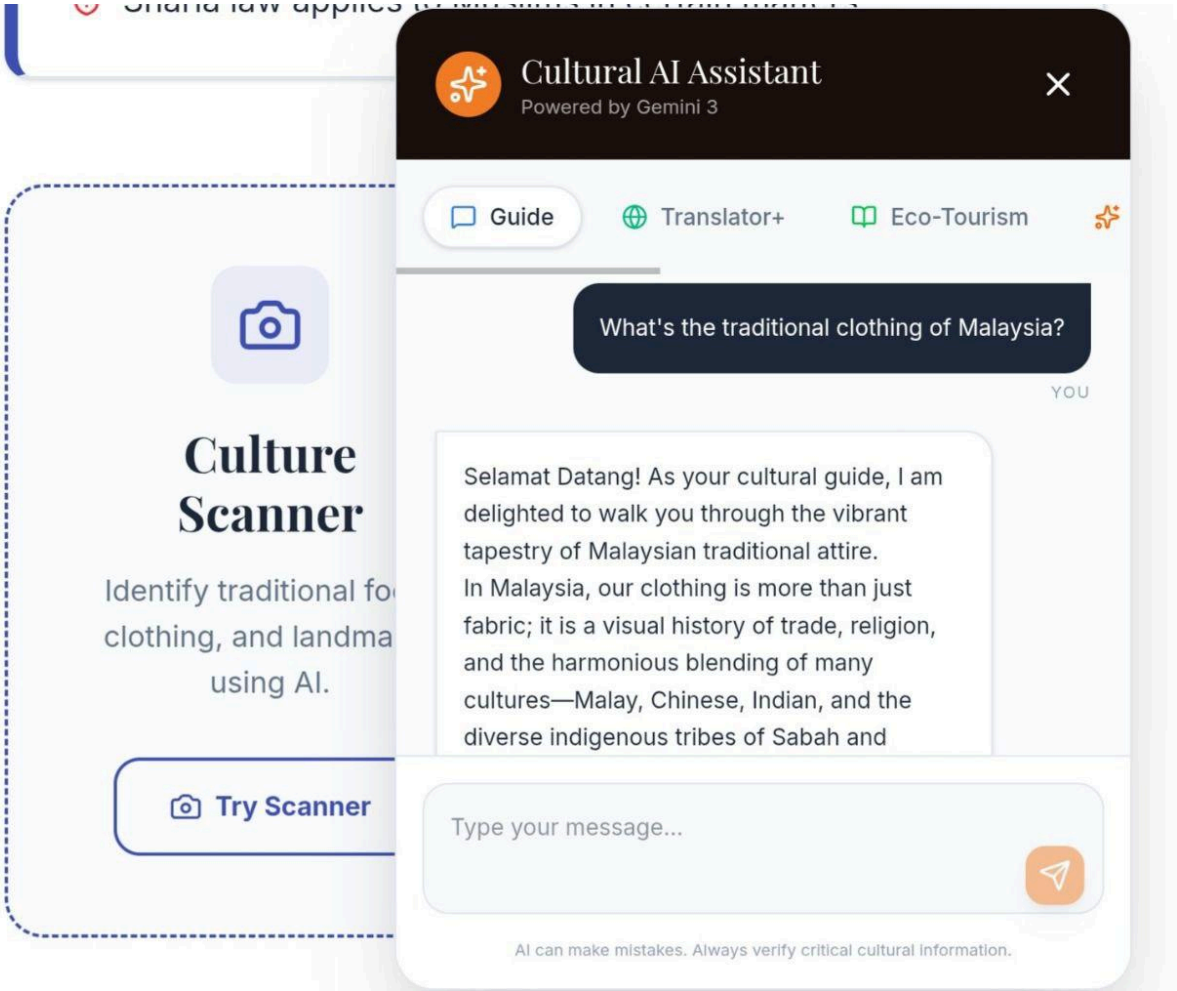

Figure 5: Prototype output successfully retrieving localised cultural data.

6.0 AI Acknowledgement

We acknowledge the use of AI tools, specifically Google AI Studio, to generate and refine the website we created as our prototype demo. In detail, Google AI Studio is used in the design and testing of our SeaAI website. The central system is the Gemini AI model by Google that helps interpret user questions to provide clear, relevant responses. However, all strategic planning, UI/UX architecture, target market identification, research on past solutions, and final factual validation were strictly human-led.

7.0 Conclusion

In conclusion, the prototype is able to demonstrate the ability to retrieve the most niche of cultural knowledge and facilitate cross-culture learning across ASEAN countries. This ensures our target market aimed towards students, business professionals, and tourists can smoothly and respectfully navigate diverse social norms, regardless of which Southeast Asian country they are visiting. Strength and weaknesses of the prototype built must be addressed to ensure SeaAI does not violate any regulations, adhere to its objectives, provide a much better user experience and to endure fast-paced market changes. Below is a listed SWOT table to efficiently show the strengths and downsides of SeaAI :

Strength	**Weakness**
● High coverage of hyper-local ASEAN cultural diversity	● Dependent on active internet connection ● Restricted to ASEAN cultures only
Opportunities	**Threats**
● Establish as the initiator as the premier, student-friendly ASEAN cultural database	● Market competition from well-funded and mainstream AI technology

Future improvements have been considered for our SeaAI to provide users optimised experience, highly accessible knowledge and practical solutions to cross-culture barriers and information.